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ECE357

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Problem Set 4

1a) If the CPU doesn't have the PCID feature, then the TLB (translation lookaside buffer) must be cleared every time by CPU the hardware invalidating the entries, which means for the new context the PTE's must be recached later. However, if the CPU does have PCID, then only the key of the cache is changed so the TLB can keep separate caches per context since it has some sort of information (the PCID) to distinguish them.

1b) Assuming the system demand pages this, the stack grows downward and expands to include the page you faulted on, and the operation is retried with a new/updated vm_area_struct

1c) We are attempting to "access memory which corresponds to just beyond the current end of file, but not beyond a page size boundary" which first gives us bytes of zero. Then, when we continue to "play" the file no longer exists since the cable was unplugged which indicates that, after we page fault, we cannot simply retry the instruction. Instead we must deliver a SIGBUS to the process which by default[1] terminates

1d) By default, the data region is created with the MAP_DENYWRITE flag, so that any kind of write *access* is blocked during file execution, with error ETXTBUSY. Even though process 555 has the same UID, and therefore all permissions, opening with read-*write mode* will fail.

1e) In a 64-bit system X86 system, it goes up to 512GB -> so from 0 up to 512GB (which is 2^39, but subtract 1 because of the 0) -> 0x 0000 007F FFFF FFFF

2)Really not sure how to get the PAs of the PGDs, ran out of time to do this, sorry.

[1] man 7

$4096_{10} = 1000_{16}$